

**ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ
TRUYỀN THÔNG**

**BÁO CÁO THỰC HÀNH LAB 05
LẬP TRÌNH HƯỚNG ĐỐI TƯỢNG
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I. Swing components

1. AWTAccumulator

```
package hust.soict.globalict.swing;
import java.awt.*;
import java.awt.event.*;
public class AWTAccumulatorHbq extends Frame {
    private TextField tfInput;
    private TextField tfOutput;
    private int sum = 0; // Accumulated sum, init to 0

    // Constructor to setup the GUI components and event handlers
    public AWTAccumulatorHbq() {
        setLayout(new GridLayout(rows:2, cols:2));
        add(new Label(text:"Enter an Integer: "));
        tfInput = new TextField(columns:10);
        add(tfInput);
        tfInput.addActionListener(new TFInputListener());
        add(new Label(text:"The Accumulated Sum is: "));
        tfOutput = new TextField(columns:10);
        tfOutput.setEditable(b:false);
        add(tfOutput);
        setTitle(title:"AWT Accumulator");
        setSize(width:350, height:120);
        setVisible(b:true);
    }

    Run | Debug
    public static void main(String[] args) {
        new AWTAccumulatorHbq();
    }

    private class TFInputListener implements ActionListener {
        @Override
        public void actionPerformed(ActionEvent evt) {
            int numberIn = Integer.parseInt(tfInput.getText());
            sum += numberIn;
            tfInput.setText(t:"");
            tfOutput.setText(sum + "");
        }
    }
}
```

Figure 1.1 Source code of AWTAccumulator

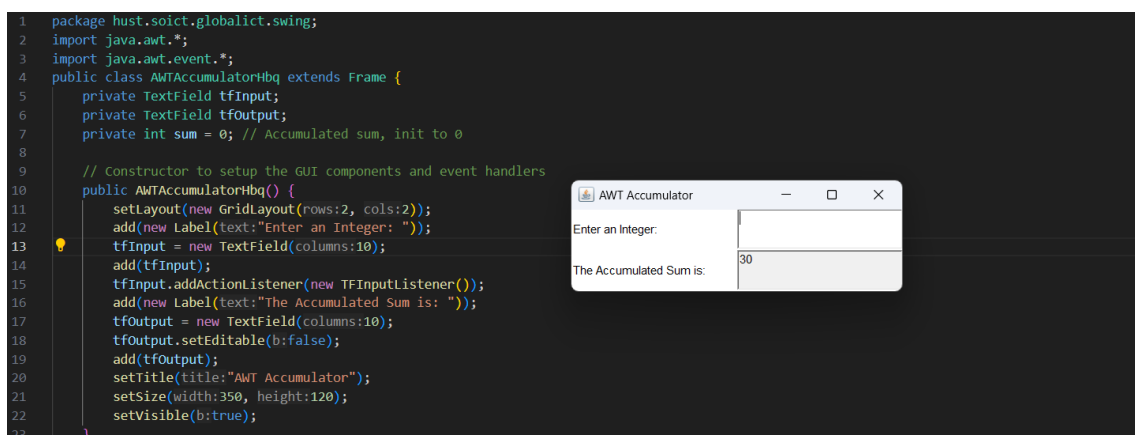


Figure 1.2 Demo of AWTAccumulator

2. SwingAccumulator

```
package hust.soict.globalict.swing;
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
public class SwingAccumulatorHbq extends JFrame {
    private JTextField tfInput;
    private JTextField tfOutput;
    private int sum = 0; // Accumulated sum, init to 0
    // Constructor to setup the GUI components and event handlers
    public SwingAccumulatorHbq() {
        Container cp = getContentPane();
        cp.setLayout(new GridLayout(rows:2, cols:2));
        cp.add(new JLabel(text:"Enter an Integer: "));
        tfInput = new JTextField(columns:10);
        cp.add(tfInput);
        tfInput.addActionListener(new TFInputListener());
        cp.add(new JLabel(text:"The Accumulated Sum is: "));
        tfOutput = new JTextField(columns:10);
        tfOutput.setEditable(b:false);
        cp.add(tfOutput);
        setTitle(title:"Swing Accumulator");
        setSize(width:350, height:120);
        setVisible(b:true);
    }
    Run | Debug
    public static void main(String[] args) {
        new SwingAccumulatorHbq();
    }
    private class TFInputListener implements ActionListener {
        @Override
        public void actionPerformed(ActionEvent evt) {
            int numberIn = Integer.parseInt(tfInput.getText());
            sum += numberIn;
            tfInput.setText(t:"");
            tfOutput.setText(sum + "");
        }
    }
}
```

Figure 1.3 Source code of SwingAccumulator

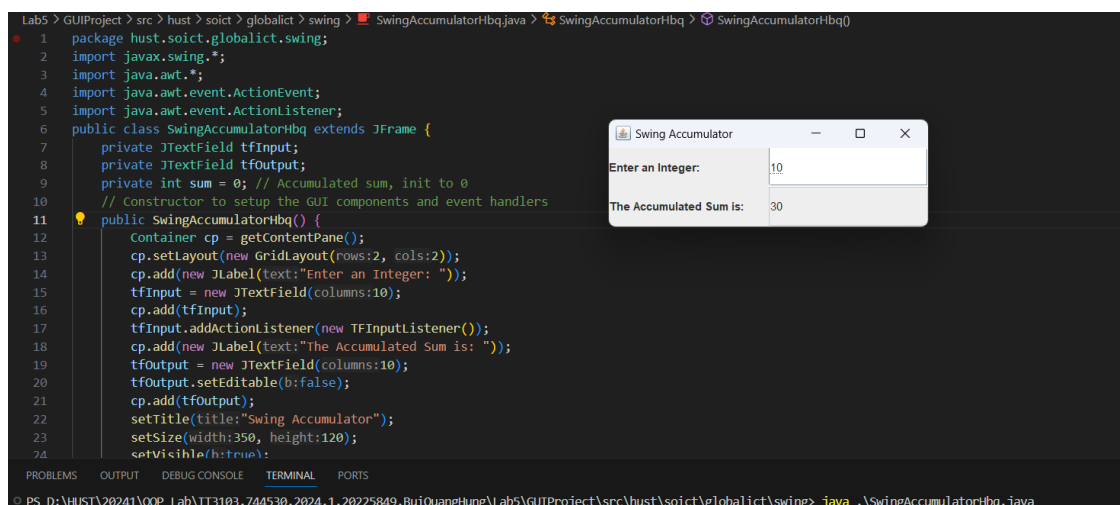


Figure 1.4 Demo of SwingAccumulator

II. Organizing Swing components with Layout Managers

1. Code

```
1 package hust.soict.globalict.swing;
2 import javax.swing.*;
3 import java.awt.*;
4 import java.awt.event.ActionEvent;
5 import java.awt.event.ActionListener;
6
7 public class NumberGridHbq extends JFrame {
8     private JButton[] btnNumbers = new JButton[10];
9     private JButton btnDelete, btnReset;
10    private JTextField tfDisplay;
11
12    public NumberGridHbq() {
13        tfDisplay = new JTextField();
14        tfDisplay.setComponentOrientation(ComponentOrientation.RIGHT_TO_LEFT);
15
16        JPanel panelButtons = new JPanel(new GridLayout(4, 3));
17        addButtons(panelButtons);
18
19        Container cp = getContentPane();
20        cp.setLayout(new BorderLayout());
21        cp.add(tfDisplay, BorderLayout.NORTH);
22        cp.add(panelButtons, BorderLayout.CENTER);
23
24        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
25        setTitle(title:"Number Grid");
26        setSize(width:200, height:200);
27        setVisible(b:true);
28    }
29
30    public static void main(String[] args) {
31        new NumberGridHbq();
32    }
33
34    void addButtons(JPanel panelButtons) {
35        ButtonListener btnListener = new ButtonListener();
36        for (int i = 1; i <= 9; i++) {
37            btnNumbers[i] = new JButton("" + i);
38            panelButtons.add(btnNumbers[i]);
39            btnNumbers[i].addActionListener(btnListener);
40        }
41    }
42}
```

Figure 2.1 Source code of NumberGrid 1

```
void addButtons(JPanel panelButtons) {
    ButtonListener btnListener = new ButtonListener();
    for (int i = 1; i <= 9; i++) {
        btnNumbers[i] = new JButton("" + i);
        panelButtons.add(btnNumbers[i]);
        btnNumbers[i].addActionListener(btnListener);
    }

    btnDelete = new JButton(text:"DEL");
    panelButtons.add(btnDelete);
    btnDelete.addActionListener(btnListener);

    btnNumbers[0] = new JButton(text:"0");
    panelButtons.add(btnNumbers[0]);
    btnNumbers[0].addActionListener(btnListener);

    btnReset = new JButton(text:"C");
    panelButtons.add(btnReset);
    btnReset.addActionListener(btnListener);
}

private class ButtonListener implements ActionListener {
    public void actionPerformed(ActionEvent e) {
        String button = e.getActionCommand();
        if(button.charAt(index:0) >= '0' && button.charAt(index:0) <= '9') {
            tfDisplay.setText(tfDisplay.getText() + button);
        }
        else if (button.equals(anObject:"DEL")) {
            //handles the "DEL" case
        }
        else {
            //handles the "C" case
        }
    }
}
```

Figure 2.2 Source code of NumberGrid 2

2. Demo

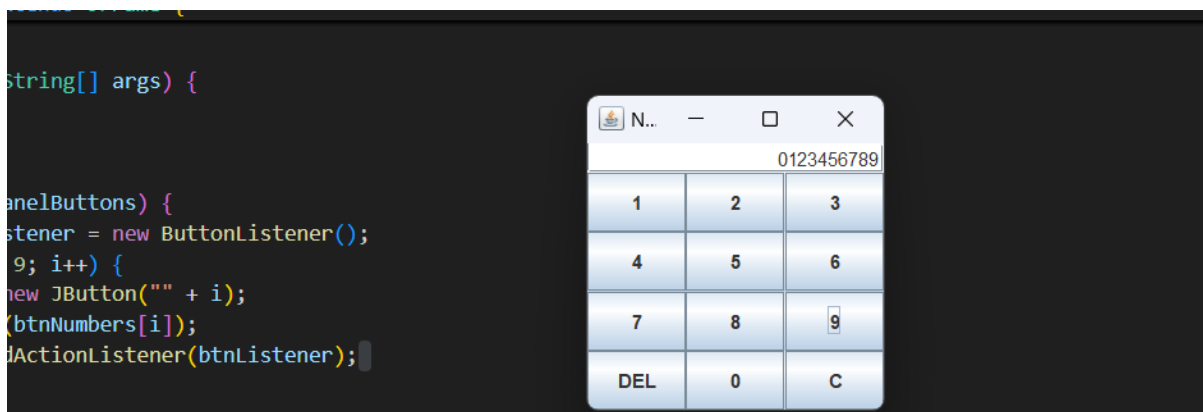


Figure 2.3 Demo buttons 0 - 9

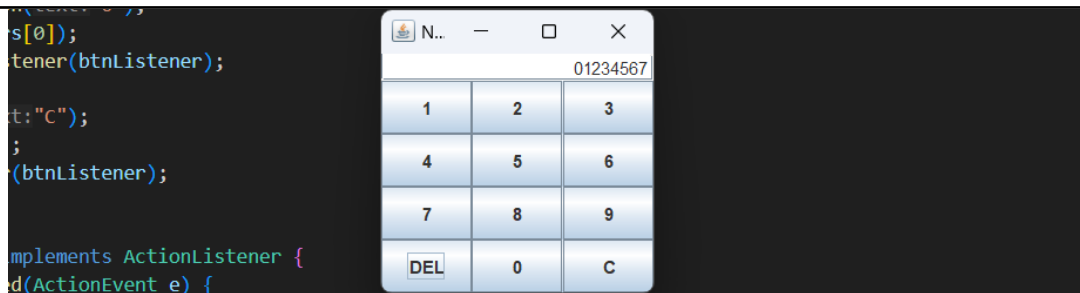


Figure 2.4 Demo DEL button

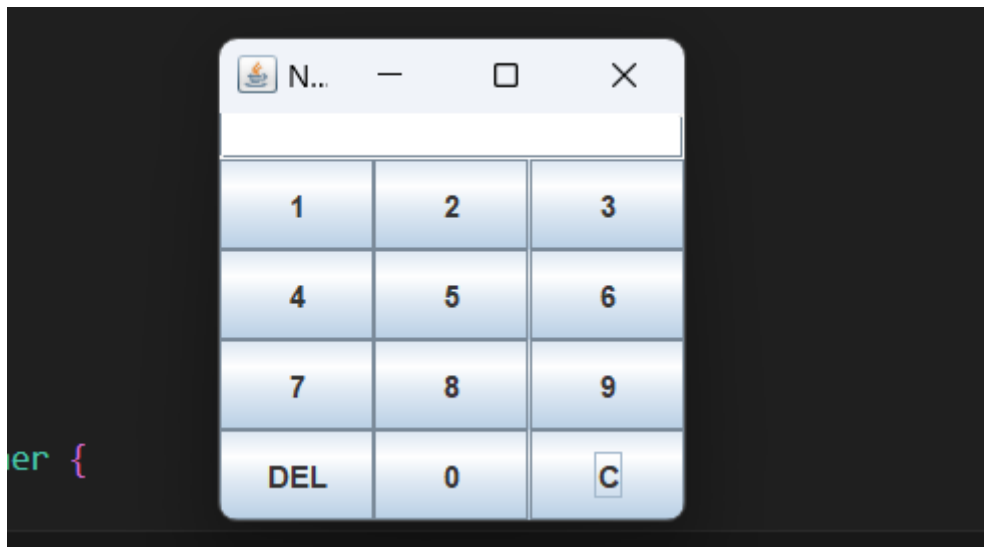


Figure 2.5 Demo C button

III. Create a graphical user interface for AIMS with Swing

1. Create class StoreScreen

```
1 package hust.soict.dsai.aims.screen;
2
3 import hust.soict.dsai.aims.media.*;
4 import java.awt.BorderLayout;
5 import java.awt.Color;
6 import java.awt.Container;
7 import java.awt.Dimension;
8 import java.awt.FlowLayout;
9 import java.awt.Font;
10 import java.awt.GridLayout;
11 import java.util.ArrayList;
12
13 import javax.swing.Box;
14 import javax.swing.BoxLayout;
15 import javax.swing.JButton;
16 import javax.swing.JFrame;
17 import javax.swing.JLabel;
18 import javax.swing.JMenu;
19 import javax.swing.JMenuBar;
20 import javax.swing.JMenuItem;
21 import javax.swing.JPanel;
22
23 import hust.soict.dsai.aims.store.StoreHungBQ;
24
25 public class StoreScreenHbq extends JFrame {
26     private static final long serialVersionUID = 1L;
27     private StoreHungBQ store;
28
29     public StoreScreenHbq(StoreHungBQ store) {
30         this.store = store;
31         Container cp = getContentPane();
32         cp.setLayout(new BorderLayout());
33
34         cp.add(createNorthHbq(), BorderLayout.NORTH);
35         cp.add(createCenter(), BorderLayout.CENTER);
36
37         setVisible(b:true);
38         setTitle(title:"Store");
39         setSize(width:1024, height:768);
```

Figure 3.1 Class StoreScreen 1

```
40
41     setVisible(b:true);
42     setTitle(title:"Store");
43     setSize(width:1024, height:768);
44 }
45
46 JPanel createNorthHbq() {
47     JPanel north = new JPanel();
48     north.setLayout(new BoxLayout(north, BoxLayout.Y_AXIS));
49     north.add(createMenuBarHbq());
50     north.add(createHeaderHbq());
51     return north;
52 }
53
54 JMenuBar createMenuBarHbq() {
55     JMenu menu = new JMenu(s:"Options");
56
57     JMenu smUpdateStore = new JMenu(s:"Update Store");
58     smUpdateStore.add(new JMenuItem(text:"Add Book"));
59     smUpdateStore.add(new JMenuItem(text:"Add CD"));
60     smUpdateStore.add(new JMenuItem(text:"Add DVD"));
61
62     menu.add(smUpdateStore);
63     menu.add(new JMenuItem(text:"View store"));
64     JMenuItem viewCartItem = new JMenuItem(text:"View cart");
65     viewCartItem.addActionListener(e -> new CartScreenHbq(cart)); // Mở giao diện Cart
66     menu.add(viewCartItem);
67
68     JMenuBar menuBar = new JMenuBar();
69     menuBar.setLayout(new FlowLayout(FlowLayout.LEFT));
70     menuBar.add(menu);
71
72     return menuBar;
73 }
74
75 JPanel createHeaderHbq() {
76     JPanel header = new JPanel();
```

Figure 3.2 Class StoreScreen 2

```
75 JPanel createHeaderHbq() {
76     JPanel header = new JPanel();
77     header.setLayout(new BorderLayout(header, BorderLayout.X_AXIS));
78
79     JLabel title = new JLabel(text:"AIMS HUNGBQ 20225849");
80     title.setFont(new Font(title.getFont().getName(), Font.PLAIN, size:50));
81     title.setForeground(Color.CYAN);
82
83     JButton cart = new JButton(text:"View cart");
84     cart.setPreferredSize(new Dimension(width:100, height:50));
85     cart.setMaximumSize(new Dimension(width:100, height:50));
86     cart.addActionListener(e -> new CartScreenHbq(this.cart)); // Mở giao diện Cart
87
88     header.add(Box.createRigidArea(new Dimension(width:10, height:10)));
89     header.add(title);
90     header.add(Box.createHorizontalGlue());
91     header.add(cart);
92     header.add(Box.createRigidArea(new Dimension(width:10, height:10)));
93
94     return header;
95 }
96
97 JPanel createCenter() {
98     JPanel center = new JPanel();
99     center.setLayout(new GridLayout(rows:0, cols:3, hgap:2, vgap:2)); // Tự động điều chỉnh số hàng dựa trên số lượng media
100
101     ArrayList<MediaHbq> mediaInStore = store.getItemsInStoreHbq();
102
103     // Duyệt qua tất cả media trong cửa hàng
104     for (int i = 0; i < mediaInStore.size(); i++) {
105         MediaStoreHbq cell = new MediaStoreHbq(mediaInStore.get(i), cart);
106         center.add(cell);
107     }
108
109     return center;
110 }
```

Figure 3.3 Class StoreScreen 3

```
112 public static void main(String[] args) {
113     // Khởi tạo Store
114     StoreHungBQ store = new StoreHungBQ();
115     CartHungBQ cart = new CartHungBQ(); // Tạo giỏ hàng
116
117     // Thêm các đối tượng Media vào cửa hàng với các lớp con cụ thể
118     MediaHbq book = new BookHbq(id:1, title:"Java Programming", category:"Book", cost:30.0f);
119     MediaHbq cd = new CompactDiscHbq(title:"Rock Music", cost:15.0f);
120     MediaHbq dvd = new DVDHungBQ(title:"Inception", category:"DVD", cost:25.0f);
121
122     // Thêm các media vào store
123     store.addMediaHbq(book);
124     store.addMediaHbq(cd);
125     store.addMediaHbq(dvd);
126
127     // Khởi tạo giao diện StoreScreen
128     StoreScreenHbq storescreen = new StoreScreenHbq(store);
129 }
130 }
131 }
```

Figure 3.4 Class StoreScreen 4

2. Create class MediaStore

```
1 package hust.soict.dsai.aims.screen;
2
3 import java.awt.Color;
4 import java.awt.FlowLayout;
5 import java.awt.Font;
6 import javax.swing.BorderFactory;
7 import javax.swing.Box;
8 import javax.swing.BoxLayout;
9 import javax.swing.JButton;
10 import javax.swing.JDialog;
11 import javax.swing.JLabel;
12 import javax.swing.JPanel;
13
14 import hust.soict.dsai.aims.cart.CartHungBQ;
15 import hust.soict.dsai.aims.media.MediaHbq;
16 import hust.soict.dsai.aims.media.PlayableHbq;
17
18 public class MediaStoreHbq extends JPanel {
19     private MediaHbq media;
20     private CartHungBQ cart;
21
22     public MediaStoreHbq(MediaHbq media, CartHungBQ cart) {
23         this.media = media;
24         this.cart = cart;
25         this.setLayout(new BoxLayout(this, BoxLayout.Y_AXIS));
26
27         JLabel title = new JLabel(media.getTitleHbq());
28         title.setFont(new Font(title.getFont().getName(), Font.PLAIN, size:20));
29         title.setAlignmentX(CENTER_ALIGNMENT);
30
31         JLabel cost = new JLabel("" + media.getCostHbq() + " $");
32         cost.setAlignmentX(CENTER_ALIGNMENT);
33
34         JPanel container = new JPanel();
35         container.setLayout(new FlowLayout(FlowLayout.CENTER));
36
37         // Nút "Add to cart"
38         JButton addToCartBtn = new JButton(text:"Add to cart");
39         addToCartBtn.addActionListener(e -> handleAddToCart());
40     }
41 }
```

Figure 3.5 Class MediaStore 1

```
39      addToCartBtn.addActionListener(e -> handleAddToCart());
40      container.add(addToCartBtn);
41      // Nút "Play" nếu media là PlayableHbq
42      if (media instanceof PlayableHbq) {
43          JButton playBtn = new JButton(text:"Play");
44          playBtn.addActionListener(e -> handlePlay());
45          container.add(playBtn);
46      }
47      this.add(Box.createVerticalGlue());
48      this.add(title);
49      this.add(cost);
50      this.add(Box.createVerticalGlue());
51      this.add(container);
52      this.add(Box.createVerticalGlue());
53      this.setBorder(BorderFactory.createLineBorder(Color.BLACK));
54  }
55  private void handleAddToCart() {
56      cart.addMediaHbq(media); // Thêm vào giỏ hàng
57      System.out.println("Added to cart: " + media.getTitleHbq());
58  }
59
60  private void handlePlay() {
61      if (media instanceof PlayableHbq) {
62          PlayableHbq playable = (PlayableHbq) media;
63          JDialog dialog = new JDialog();
64          dialog.setTitle(title:"Playing Media");
65          dialog.setSize(width:300, height:200);
66          dialog.setLayout(new FlowLayout());
67
68          JLabel label = new JLabel("Playing: " + media.getTitleHbq());
69          label.setFont(new Font(label.getFont().getName(), Font.PLAIN, size:16));
70          dialog.add(label);
71
72          dialog.setVisible(b:true);
73      }
74  }
75  }
```

Figure 3.6 Class MediaStore 2

3. Demo

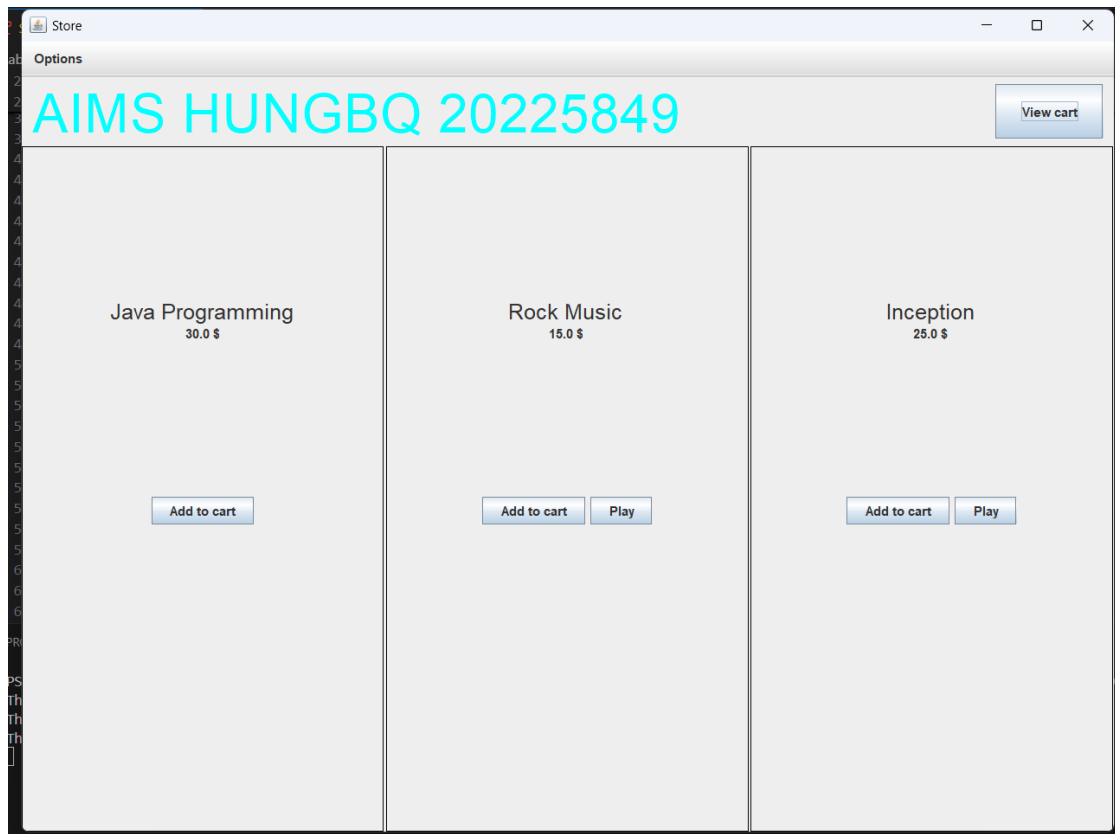


Figure 3.7 StoreScreen

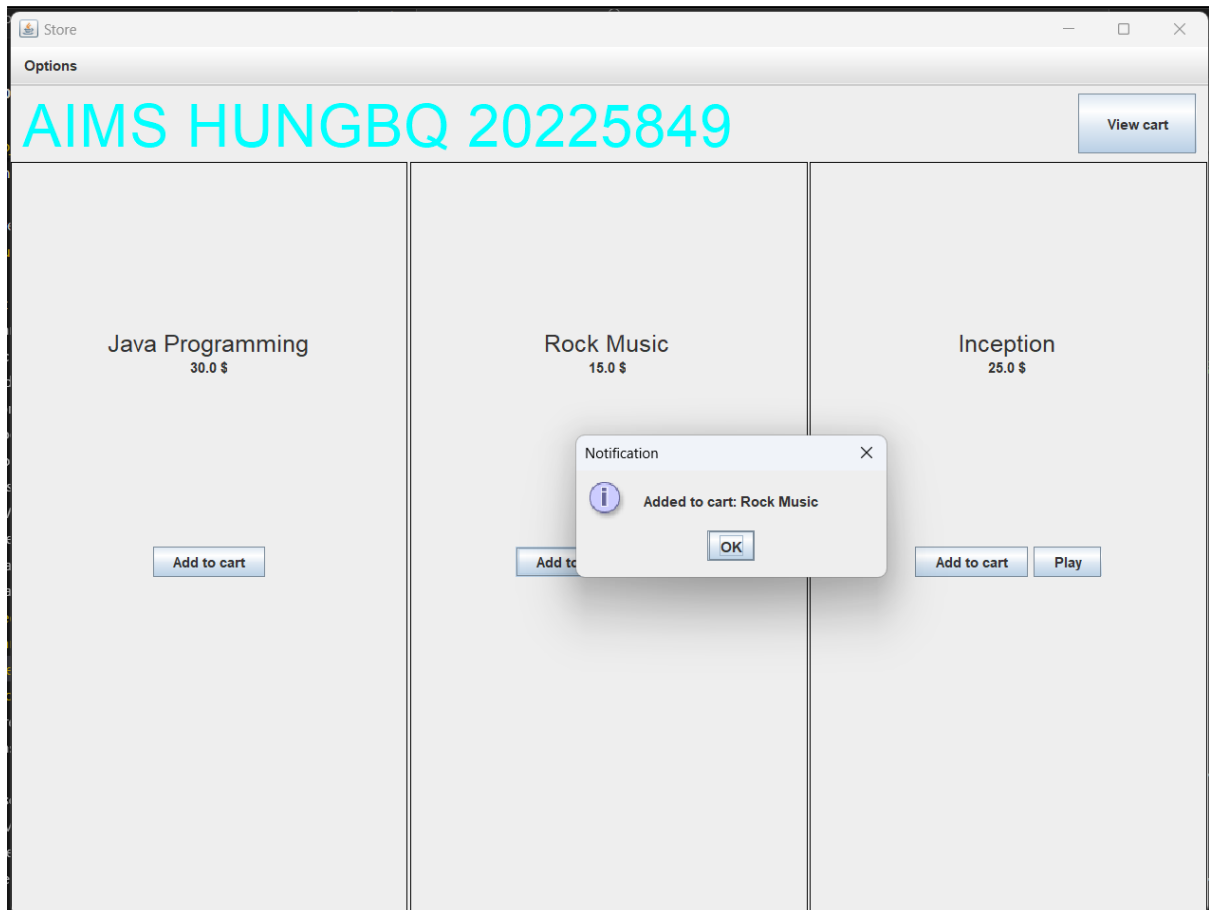


Figure 3.8 Demo Add to cart button

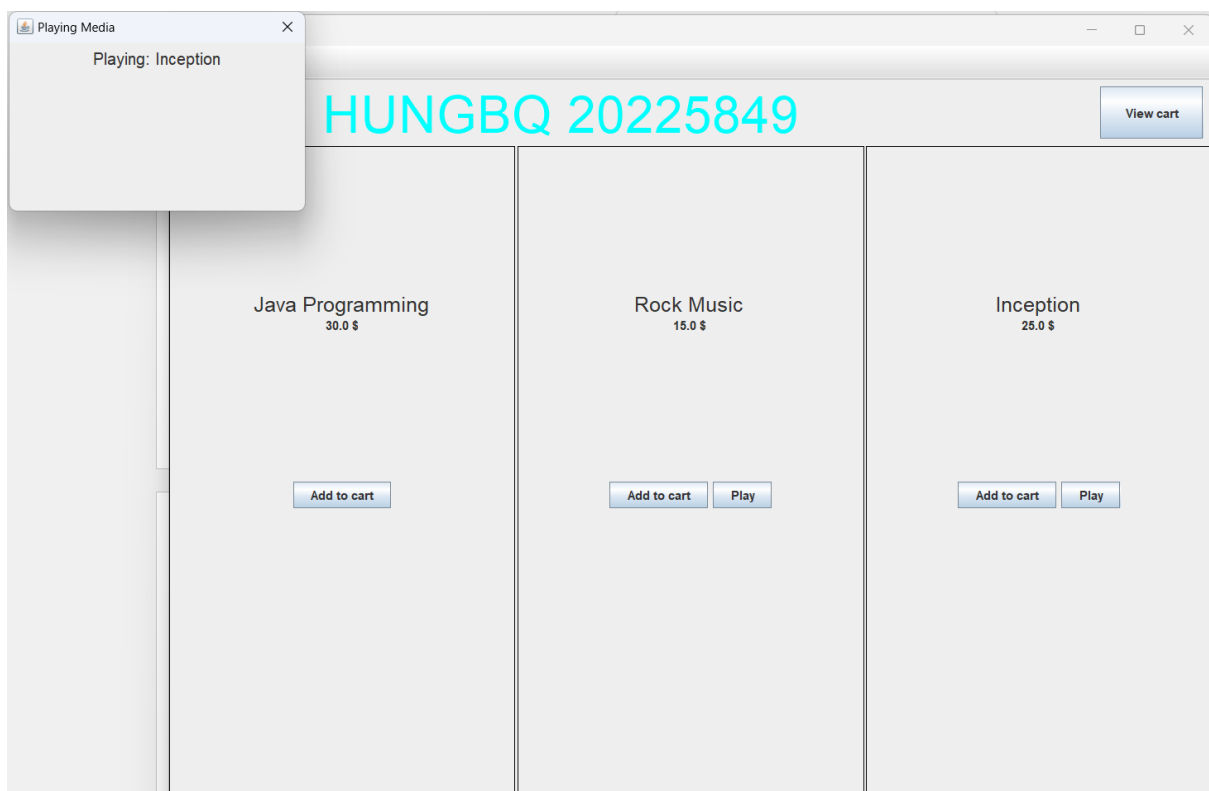


Figure 3.9 Demo Play button

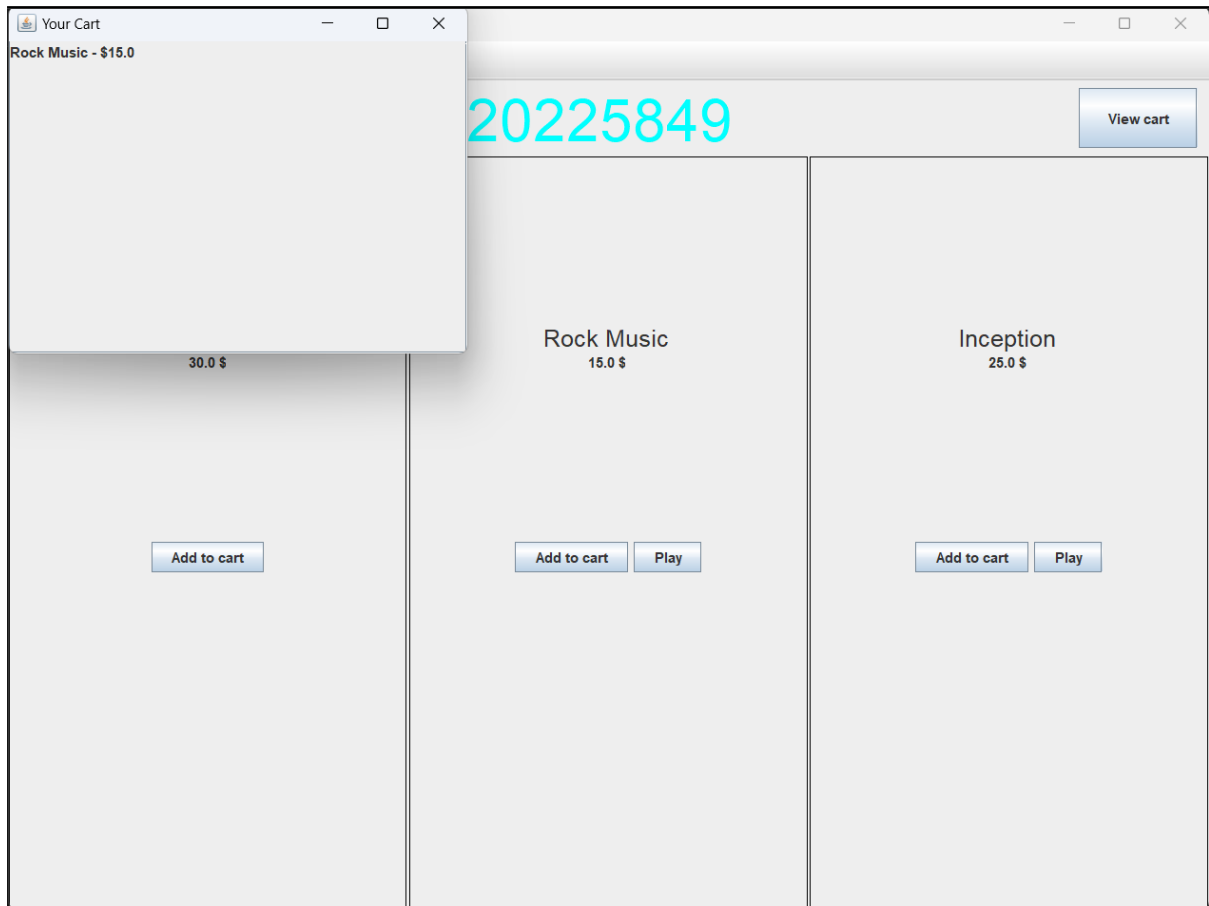


Figure 3.10 Demo View cart button

IV. JavaFX API

1. Create class Painter

```
package hust.soict.dsai.javafx;

import javafx.application.Application;
import javafx.fxml.FXMLLoader;
import javafx.scene.Parent;
import javafx.scene.Scene;
import javafx.stage.Stage;

public class PainterHbq extends Application{
    @Override
    public void start(Stage stage) throws Exception {
        Parent root = FXMLLoader.load(getClass().getResource("name: \"/hust/soict/dsai/javafx/PainterHbq.fxml\""));
        Scene scene = new Scene(root);
        stage.setTitle("Painter");
        stage.setScene(scene);
        stage.show();
    }
    > public static void main(String[] args) { launch(args); }
}
```

Figure 4.1 Class Painter

2. Create Painter.fxml

```
package hust.soict.dsai.javafx;

import javafx.application.Application;
import javafx.fxml.FXMLLoader;
import javafx.scene.Parent;
import javafx.scene.Scene;
import javafx.stage.Stage;

public class PainterHbq extends Application{
    @Override
    public void start(Stage stage) throws Exception {
        Parent root = FXMLLoader.load(getClass().getResource("/hust/soict/dsai/javafx/PainterHbq.fxml"));
        Scene scene = new Scene(root);
        stage.setTitle("Painter");
        stage.setScene(scene);
        stage.show();
    }
    public static void main(String[] args) { launch(args); }
}
```

Figure 4.2 Painter.fxml 1

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <?import javafx.geometry.Insets?>
4 <?import javafx.scene.control.Button?>
5 <?import javafx.scene.control.RadioButton?>
6 <?import javafx.scene.control.TitledPane?>
7 <?import javafx.scene.control.ToggleGroup?>
8 <?import javafx.scene.layout.AnchorPane?>
9 <?import javafx.scene.layout.BorderPane?>
10 <?import javafx.scene.layout.Pane?>
11 <?import javafx.scene.layout.VBox?>
12 <?import javafx.scene.text.Font?>
13
14 <BorderPane maxHeight="-Infinity" maxWidth="-Infinity" minHeight="-Infinity" minWidth="-Infinity"
15     prefHeight="480.0" prefWidth="640.0" xmlns="http://javafx.com/javafx/23" xmlns:fx="http://javafx.com/fxml/1"
16     fx:controller="hust.soict.dsai.javafx.PainterControllerHbq">
17     <padding>
18         <Insets bottom="8.0" left="8.0" right="8.0" top="8.0" />
19     </padding>
20     <Left>
21         <VBox maxHeight="1.7976931348623157E308" prefHeight="465.0" prefWidth="78.0" spacing="8.0" BorderPane.alignment="CENTER">
22             <BorderPane.margin>
23                 <Insets right="8.0" />
24             </BorderPane.margin>
25             <children>
26                 <TitledPane animated="false" maxWidth="1.7976931348623157E308" prefHeight="86.0" prefWidth="70.0" text="Tools">
27                     <content>
28                         <AnchorPane minHeight="0.0" minWidth="0.0" prefHeight="104.0" prefWidth="76.0">
29                             <children>
30                                 <RadioButton layoutX="27.0" layoutY="5.0" mnemonicParsing="false" onAction="#penModeActivated" text="Pen"
31                                     AnchorPane.leftAnchor="0.0" AnchorPane.topAnchor="0.0">
32                                     <toggleGroup>
33                                         <ToggleGroup fx:id="ToolOption" />
34                                     </toggleGroup>
35                                 </RadioButton>
36                             </children>
37                         </AnchorPane>
38                     </content>
39                 </TitledPane>
40                 <Button maxHeight="1.7976931348623157E308" mnemonicParsing="false" onAction="#clearButtonPressed" text="Clear">
41                     <font>
42                         <font size="13.0" />
43                     </font>
44                 </Button>
45             </children>
46         </VBox>
47     </Left>
48     <center>
49         <Pane fx:id="drawingAreaPane" onMouseDragged="#drawingAreaMouseDragged" onMousePressed="#drawingAreaMouseDragged" prefHeight="200.0"
50             prefWidth="200.0" style="-fx-background-color: white;" BorderPane.alignment="CENTER" />
51     </center>
52 </BorderPane>
```

Figure 4.3 Painter.fxml 2

```
30 <RadioButton layoutX="27.0" layoutY="5.0" mnemonicParsing="false" onAction="#penModeActivated" text="Pen"
31     AnchorPane.leftAnchor="0.0" AnchorPane.topAnchor="0.0">
32     <toggleGroup>
33         <ToggleGroup fx:id="ToolOption" />
34     </toggleGroup>
35 </RadioButton>
36 </children>
37 </AnchorPane>
38 </content>
39 </TitledPane>
40 <Button maxHeight="1.7976931348623157E308" mnemonicParsing="false" onAction="#clearButtonPressed" text="Clear">
41     <font>
42         <font size="13.0" />
43     </font>
44 </Button>
45 </children>
46 </VBox>
47 </Left>
48 <center>
49     <Pane fx:id="drawingAreaPane" onMouseDragged="#drawingAreaMouseDragged" onMousePressed="#drawingAreaMouseDragged" prefHeight="200.0"
50         prefWidth="200.0" style="-fx-background-color: white;" BorderPane.alignment="CENTER" />
51 </center>
52 </BorderPane>
```

Figure 4.4 Painter.fxml 3

3. Create class PainterController

```
1 package hust.soict.dsai.javafx;
2
3 import javafx.event.ActionEvent;
4 import javafx.fxml.FXML;
5 import javafx.scene.input.MouseEvent;
6 import javafx.scene.layout.*;
7 import javafx.scene.paint.Color;
8 import javafx.scene.shape.Circle;
9
10 public class PainterControllerHbq {
11     private Color pen; // 3 usages
12     @FXML
13     private Pane drawingAreaPane;
14
15     @FXML
16     void clearButtonPressed(ActionEvent event) { drawingAreaPane.getChildren().clear(); }
17
18     @FXML
19     void drawingAreaMouseDragged(MouseEvent event) {
20         Circle newCircle = new Circle(event.getX(), event.getY(), 4, pen);
21         drawingAreaPane.getChildren().add(newCircle);
22     }
23
24     @FXML
25     void penModeActivated(ActionEvent event) {
26         pen = Color.BLACK;
27     }
28
29     @FXML
30     void eraserModeActivated(ActionEvent event) {
31         pen = Color.WHITE;
32     }
33 }
34 }
```

Figure 4.5 PainterController

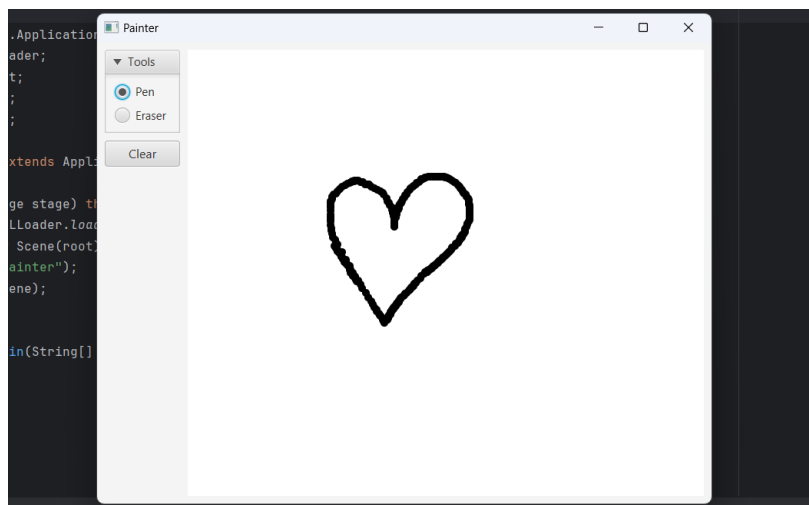


Figure 4.6 Use Pen

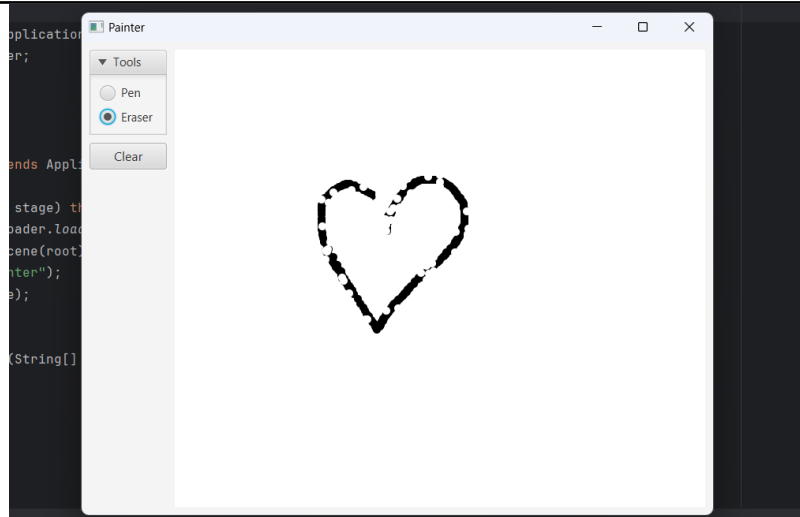


Figure 4.6 Use Eraser

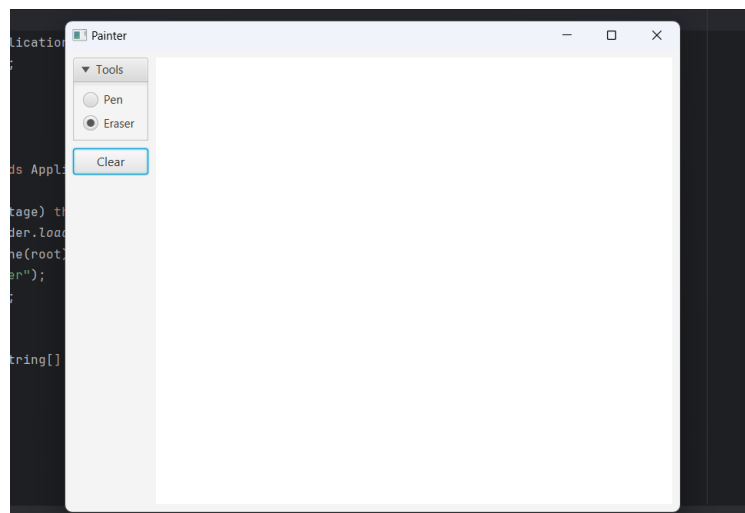


Figure 4.7 Clear Button

V. View Cart Screen

1. Create Cart.fxml

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <?import javafx.geometry.Insets?>
4 <?import javafx.scene.control.Button?>
5 <?import javafx.scene.control.ButtonBar?>
6 <?import javafx.scene.control.Label?>
7 <?import javafx.scene.control.Menu?>
8 <?import javafx.scene.control.MenuBar?>
9 <?import javafx.scene.control.MenuItem?>
10 <?import javafx.scene.control.RadioButton?>
11 <?import javafx.scene.control.TableColumn?>
12 <?import javafx.scene.control.TableView?>
13 <?import javafx.scene.control.TextField?>
14 <?import javafx.scene.control.ToggleGroup?>
15 <?import javafx.scene.layout.BorderPane?>
16 <?import javafx.scene.layout.HBox?>
17 <?import javafx.scene.layout.VBox?>
18 <?import javafx.scene.text.Font?>
19
20 <BorderPane maxHeight="-Infinity" maxWidth="-Infinity" minHeight="-Infinity" minWidth="-Infinity" prefHeight="768.0"
21 prefWidth="1024.0" xmlns="http://javafx.com/javafx/23.0.1" xmlns:fx="http://javafx.com/fxml/1">
22 <top>
23 <VBox prefWidth="328.0" BorderPane.alignment="CENTER">
24 <children>
25 <MenuBar prefHeight="25.0" prefWidth="55.0">
26 <menus>
27 <Menu mnemonicParsing="false" text="Options">
28 <items>
29 <Menu mnemonicParsing="false" text="Update Store">
30 <items>
31 <MenuItem mnemonicParsing="false" text="Add Book" />
```

Figure 5.1 Cart.fxml 1

```
31 <MenuItem mnemonicParsing="false" text="Add Book" />
32 <MenuItem mnemonicParsing="false" text="Add CD" />
33 <MenuItem mnemonicParsing="false" text="Add DVD" />
34 </items>
35 </Menu>
36 <MenuItem mnemonicParsing="false" text="View Sotre" />
37 <MenuItem mnemonicParsing="false" text="View Cart" />
38 </items>
39 </Menu>
40 </menus>
41 </MenuBar>
42 <Label text="CART" textFill="#ff00d3">
43 <font>
44 <Font size="50.0" />
45 </font>
46 <padding>
47 <Insets right="10.0" />
48 </padding>
49 </Label>
50 </children>
51 <padding>
52 <Insets right="10.0" />
53 </padding>
54 </VBox>
55 </top>
```

Figure 5.2 Cart.fxml 2

```
56 <center>
57 <VBox prefHeight="200.0" prefWidth="100.0" BorderPane.alignment="CENTER">
58 <padding>
59 <Insets right="10.0" />
60 </padding>
61 <children>
62 <HBox alignment="CENTER_LEFT" prefWidth="200.0" spacing="10.0">
63 <padding>
64 <Insets bottom="10.0" top="10.0" />
65 </padding>
66 <children>
67 <Label text="Filter" />
68 <TextField fx:id="tfFilter" />
69 <RadioButton fx:id="radioBtnFilterId" mnemonicParsing="false" selected="true" text="By ID">
70 <toggleGroup>
71 <ToggleGroup fx:id="filterCategory" />
72 </toggleGroup>
73 </RadioButton>
74 <RadioButton fx:id="radioBtnFilterTitle" mnemonicParsing="false" text="By Title" toggleGroup="$filterCategory" />
75 </children>
76 <VBox.margin>
77 <Insets />
78 </VBox.margin>
79 </HBox>
80 <TableView fx:id="tblMedia">
81 <columns>
82 <TableColumn fx:id="colMediaTitle" prefWidth="75.0" text="Title" />
83 <TableColumn fx:id="colMediaCategory" prefWidth="75.0" text="Category" />
84 <TableColumn fx:id="colMediaCost" prefWidth="75.0" text="Cost" />
85 </columns>
```

Figure 5.3 Cart.fxml 3

```
86 </columns>
87 <columnResizePolicy>
88 <TableView fx:constant="CONSTRAINED_RESIZE_POLICY" />
89 </columnResizePolicy>
90 </TableView>
91 <ButtonBar prefHeight="40.0" prefWidth="200.0">
92 <buttons>
93 <Button fx:id="btnPlay" mnemonicParsing="false" text="Play" />
94 <Button fx:id="btnRemove" mnemonicParsing="false" onAction="#btnRemovePressed" text="Remove" />
95 </buttons>
96 </ButtonBar>
97 <children>
98 <BorderPane.margin>
99 <Insets left="5.0" />
100 </BorderPane.margin>
101 </VBox>
102 </center>
103 <right>
104 <VBox alignment="TOP_CENTER" prefHeight="200.0" BorderPane.alignment="CENTER">
105 <padding>
106 <Insets top="50.0" />
107 </padding>
108 <children>
109 <HBox alignment="CENTER">
110 <children>
```

Figure 5.4 Cart.fxml 4

```
108 <HBox alignment="CENTER">
109 <children>
110 <Label lineSpacing="10.0" text="Total: ">
111 <font>
112 <Font size="24.0" />
113 </font>
114 </Label>
115 <Label text="0 $" textFill="RED">
116 <font>
117 <Font size="24.0" />
118 </font>
119 </Label>
120 </children>
121 </HBox>
122 <Button mnemonicParsing="false" style="-fx-background-color: ff00d3;" text="Place Order" textFill="WHITE">
123 <font>
124 <Font size="24.0" />
125 </font>
126 <VBox.margin>
127 <Insets right="5.0" />
128 </VBox.margin>
129 </Button>
130 </children>
131 </VBox>
132 </right>
133 </BorderPane>
```

Figure 5.5 Cart.fxml

2. Create class CartScreen

```
1 package hust.soict.dsai.aims.screen.cart;
2 import java.io.IOException;
3 import javax.swing.JFrame;
4 import hust.soict.dsai.aims.cart.CartHungBQ;
5 import javafx.application.Platform;
6 import javafx.embed.swing.JFXPanel;
7 import javafx.fxml.FXMLLoader;
8 import javafx.scene.Parent;
9 import javafx.scene.Scene;
10 public class CartScreenHbq extends JFrame{
11     private CartHungBQ cart; 1 usage
12     public CartScreenHbq(CartHungBQ cart) { 1 usage
13         super();
14         this.cart = cart;
15         JFXPanel fxPanel = new JFXPanel();
16         this.add(fxPanel);
17         this.setTitle("Cart");
18         this.setVisible(true);
19         Platform.runLater(new Runnable() {
20             @Override
21             public void run() {
22                 try {
23                     FXMLLoader loader = new FXMLLoader(getClass().getResource("name: "/hust/soict/dsai/aims/screen/cart/CartHbq.fxml"));
24                     CartScreenControllerHbq controller = new CartScreenControllerHbq(cart);
25                     loader.setController(controller);
26                     Parent root = loader.load();
27                     fxPanel.setScene(new Scene(root));
28                 } catch (IOException e) {
29                     e.printStackTrace();
30                 }
31             }
32         });
33     }
34 }
```

Figure 5.6 CartScreen class

3. Create CartScreenController

```
1 package hust.soict.dsai.aims.screen.cart;
2 import hust.soict.dsai.aims.cart.CartHungBQ;
3 import hust.soict.dsai.aims.media.MediaHbq;
4 import javafx.fxml.FXML;
5 import javafx.scene.control.TableColumn;
6 import javafx.scene.control.TableView;
7 import javafx.scene.control.ToggleGroup;
8 import javafx.scene.control.cell.PropertyValueFactory;
9 public class CartScreenControllerHbq {
10     private CartHungBQ cart; 2 usages
11     @FXML 1 usage
12     private TableColumn<MediaHbq, String> colMediaCategory;
13     @FXML 1 usage
14     private TableColumn<MediaHbq, Float> colMediaCost;
15     @FXML 1 usage
16     private TableColumn<MediaHbq, String> colMediaTitle;
17     @FXML no usages
18     private ToggleGroup filterCategory;
19     @FXML 1 usage
20     private TableView<MediaHbq> tblMedia;
21
22     public CartScreenControllerHbq (CartHungBQ cart) { 1 usage
23         super();
24         this.cart = cart;
25     }
26     @FXML no usages
```

Figure 5.7 CartScreenController class 1

```
}  
@FXML no usages  
private void initialize() {  
    colMediaTitle.setCellValueFactory(new PropertyValueFactory<MediaHbq, String>(s: "Title"));  
    colMediaCategory.setCellValueFactory(new PropertyValueFactory<MediaHbq, String>(s: "Category"));  
    colMediaCost.setCellValueFactory(new PropertyValueFactory<MediaHbq, Float>(s: "Cost"));  
    tblMedia.setItems(this.cart.getItemsHbq());  
}  
}
```

Figure 5.8 CartScreenController class 2

4. Demo

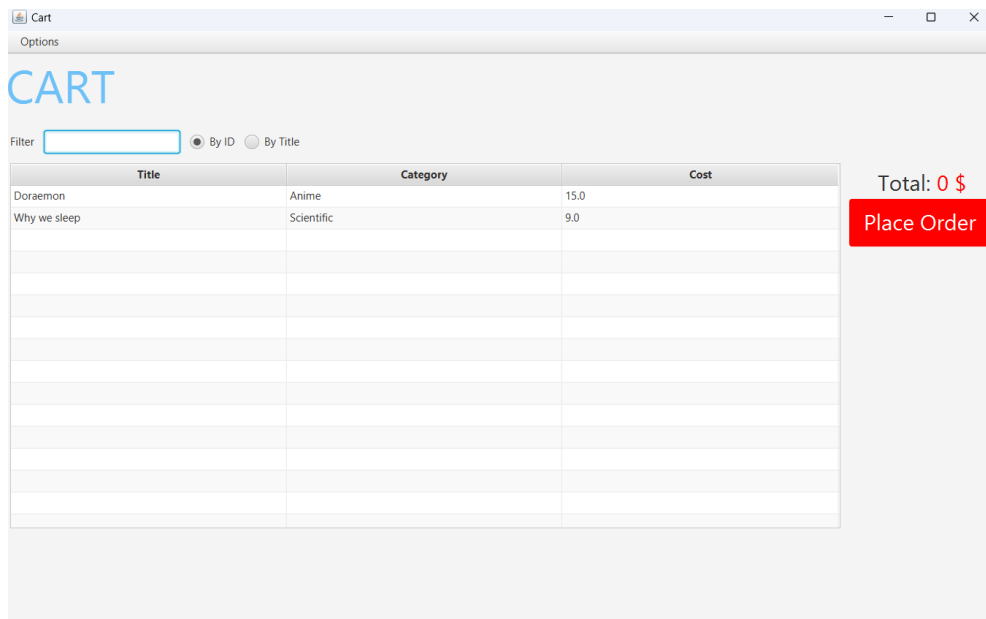


Figure 5.9 CartScreen Demo

VI. Updating buttons based on selected item in TableView – ChangeListener

1. Edit class CartScreenController

```
1 package hust.soict.dsai.aims.screen.cart;
2 import hust.soict.dsai.aims.cart.CartHungBQ;
3 import hust.soict.dsai.aims.media.MediaHbq;
4 import hust.soict.dsai.aims.media.PlayableHbq;
5 import javafx.beans.property.SimpleObjectProperty;
6 import javafx.beans.property.SimpleStringProperty;
7 import javafx.beans.value.ChangeListener;
8 import javafx.beans.value.ObservableValue;
9 import javafx.collections.FXCollections;
10 import javafx.event.ActionEvent;
11 import javafx.fxml.FXML;
12 import javafx.scene.control.Button;
13 import javafx.scene.control.RadioButton;
14 import javafx.scene.control.TableColumn;
15 import javafx.scene.control.TableView;
16 import javafx.scene.control.TextField;
17 import javafx.scene.control.ToggleGroup;
18 public class CartScreenControllerHbq {
19     private CartHungBQ cart;
20     @FXML 3 usages
21     private Button btnPlay;
22     @FXML 2 usages
23     private Button btnRemove;
24     @FXML 1 usage
25     private TableColumn<MediaHbq, String> colMediaCategory;
26     @FXML 1 usage
27     private TableColumn<MediaHbq, Float> colMediaCost;
28     @FXML 1 usage
29     private TableColumn<MediaHbq, String> colMediaTitle;
30     @FXML no usages
31     private ToggleGroup filterCategory;
32     @FXML 3 usages
33     private TableView<MediaHbq> tblMedia;
34     @FXML no usages
```

Figure 6.1 CartScreenController 1

```
35 private RadioButton radioBtnFilterId;
36 @FXML no usages
37 private RadioButton radioBtnFilterTitle;
38 @FXML 1 usage
39 private TextField tfFilter;
40 public CartScreenControllerHbq (CartHungBQ cart) { 2 usages
41     super();
42     this.cart = cart;
43 }
44 @FXML no usages
45 private void initialize() {
46     colMediaTitle.setCellValueFactory( CellDataFeatures<MediaHbq, String> cellData -> new SimpleStringProperty(cellData.getValue().getTitleHbq()));
47     colMediaCategory.setCellValueFactory( CellDataFeatures<MediaHbq, String> cellData -> new SimpleStringProperty(cellData.getValue().getCategoryHbq()));
48     colMediaCost.setCellValueFactory( CellDataFeatures<MediaHbq, Float> cellData -> new SimpleObjectProperty<>(cellData.getValue().getCostHbq()));
49     tblMedia.setItems(FXCollections.observableArrayList(cart.getItemsHbq()));
50     btnPlay.setVisible(true);
51     btnRemove.setVisible(true);
52     tblMedia.getSelectionModel().selectedItemProperty().addListener(
53         new ChangeListener<MediaHbq>() {
54             @Override
55             public void changed(ObservableValue<? extends MediaHbq> observable, MediaHbq oldValue, MediaHbq newValue) {
56                 if (newValue != null) {
57                     updateButtonBar(newValue);
58                 }
59             }
60         });
61     tfFilter.textProperty().addListener(new ChangeListener<String>() {
62         @Override
63         public void changed(ObservableValue<? extends String> observable, String oldValue, String newValue) {
64             showFilterMedia(newValue);
65         }
66     });
67 }
```

Figure 6.2 CartScreenController 2

```
68 }
69 }
70 void updateButtonBar(MediaHbq media) { 1 usage
71     btnRemove.setVisible(true);
72     if(media instanceof PlayableHbq) btnPlay.setVisible(true);
73     else btnPlay.setVisible(false);
74 }
75 void showFilterMedia(String s) { 1 usage
76     cart.searchByTitleHbq(s);
77 }
78 @FXML no usages
79 void btnRemovePressed(ActionEvent event) {
80     MediaHbq media = tblMedia.getSelectionModel().getSelectedItem();
81     cart.removeMediaHbq(media);
82 }
83 }
```

Figure 6.3 CartScreenController 3

2. Demo

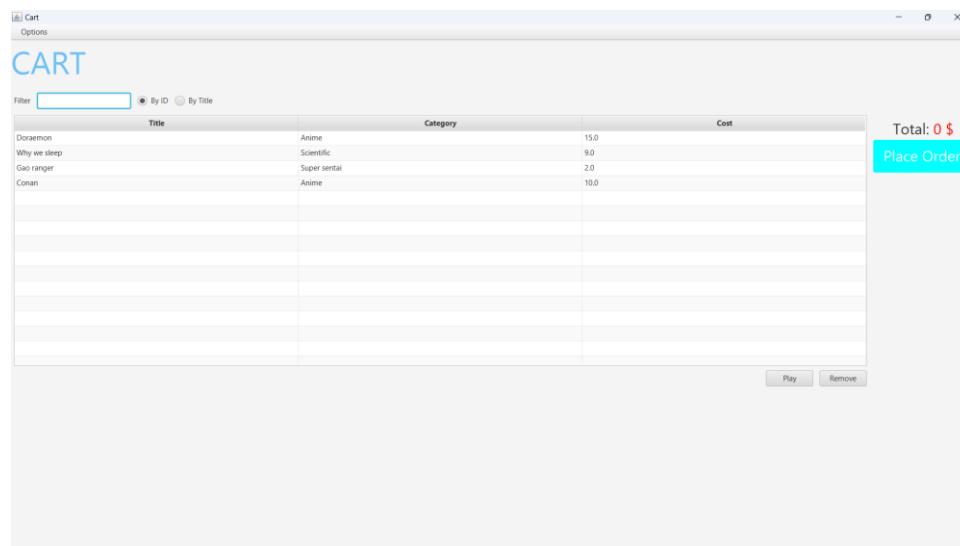


Figure 6.4 Demo media playable

VII. Deleting a media

1. Code

```
85  @FXML no usages
86  private void btnRemovePressed(ActionEvent event) {
87      MediaHbq media = tblMedia.getSelectionModel().getSelectedItem();
88      if (media != null) {
89          cart.removeMediaHbq(media);
90          // Refresh the TableView to reflect the updated cart
91          tblMedia.setItems(FXCollections.observableArrayList(cart.getItemsHbq()));
92      }
93  }
94
95  }
```

Figure 7.1 btnRemovePressed Method

2. Demo

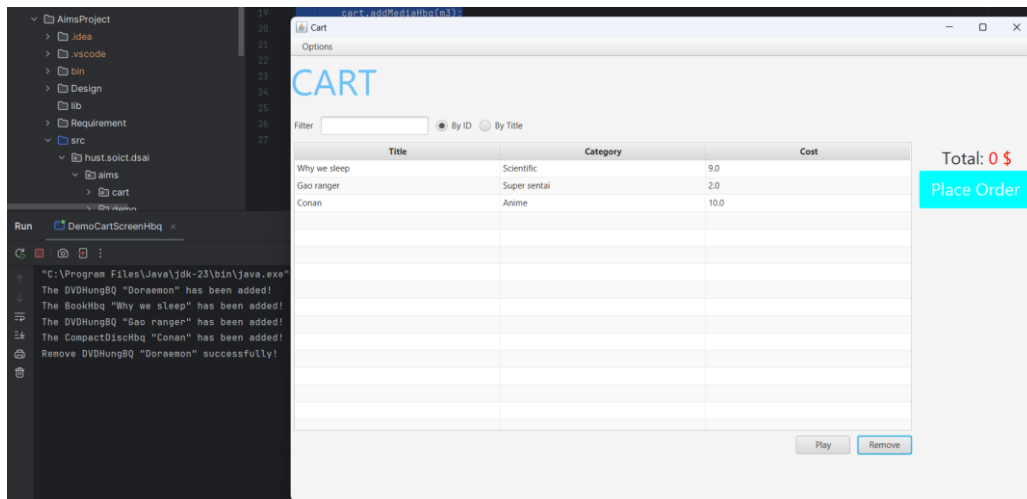


Figure 7.2 button Remove

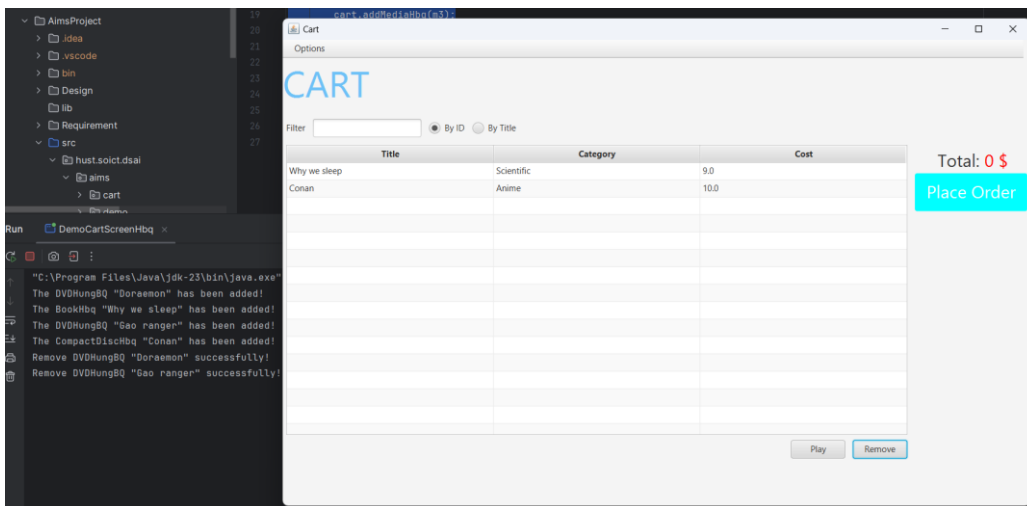


Figure 7.3 button Remove

VIII. Complete the Aims GUI Application

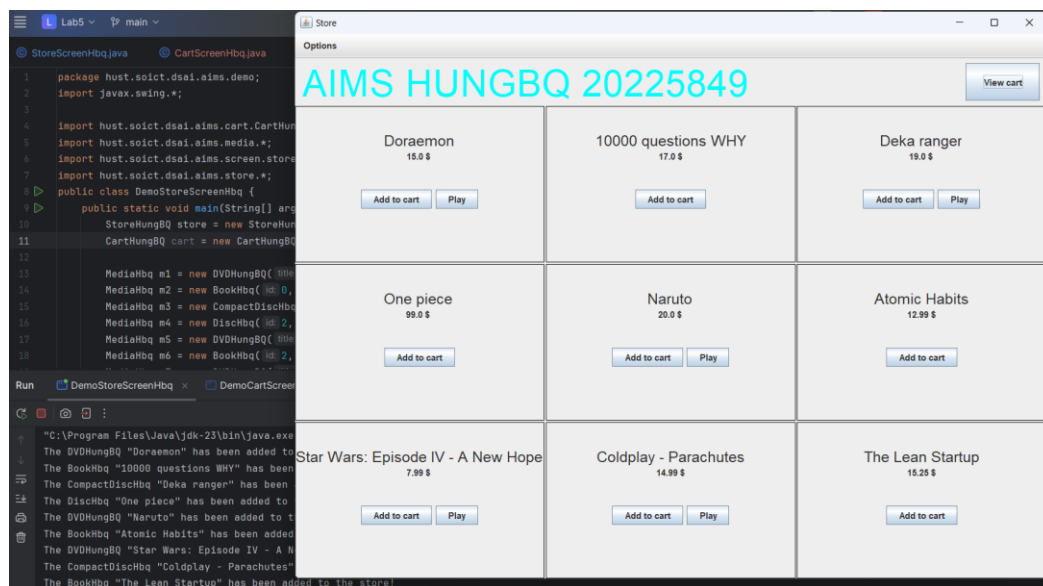


Figure 8.1 Store before add book

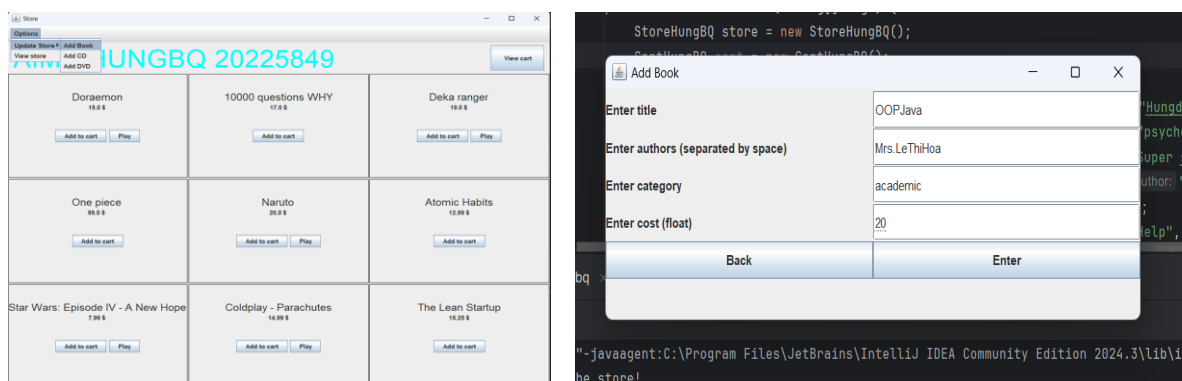


Figure 8.2 Add book

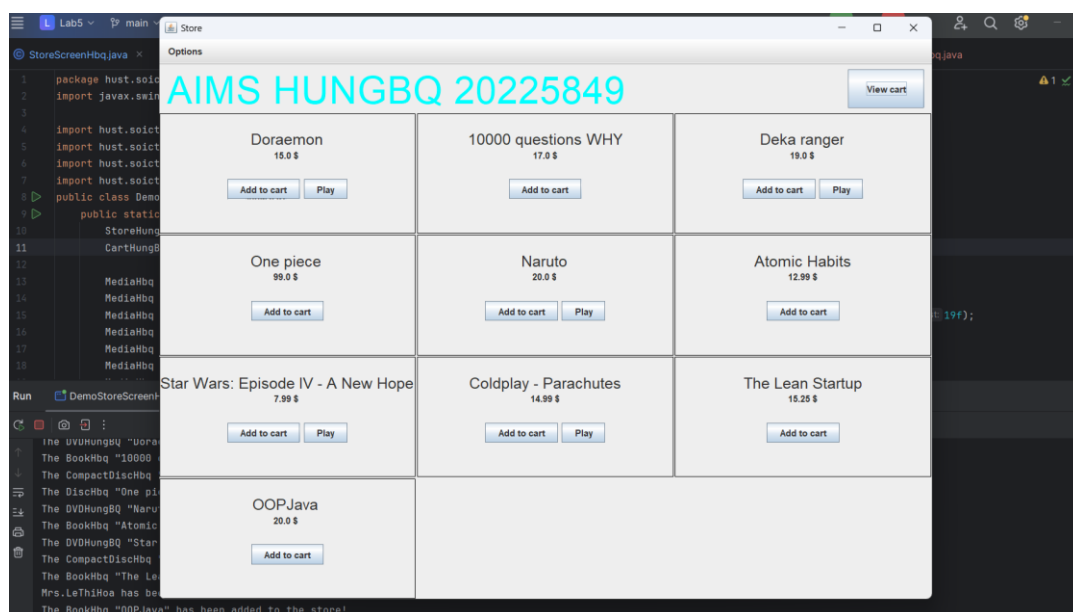


Figure 8.3 Store after add book

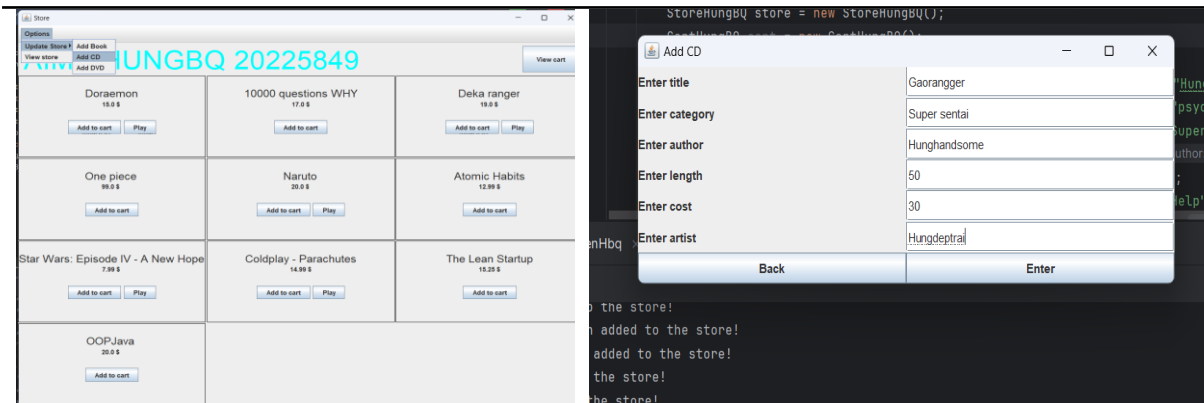


Figure 8.4 Add CD



Figure 8.5 Store after add CD

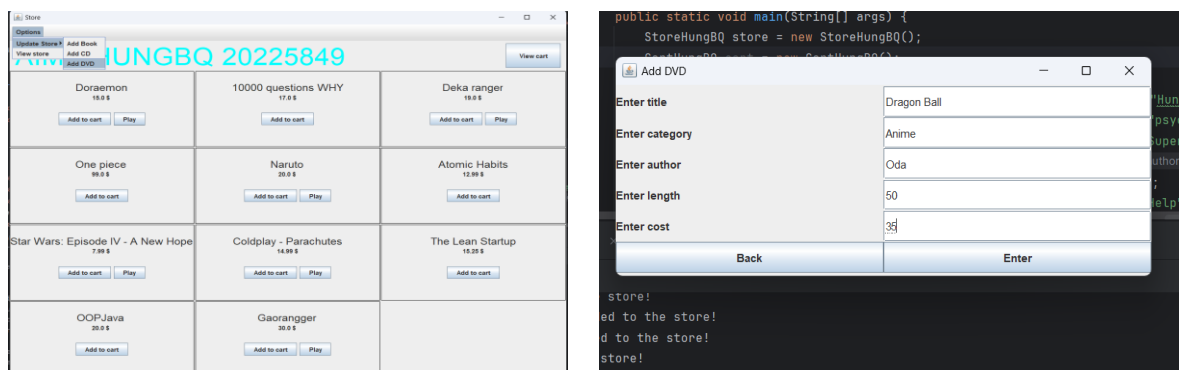


Figure 8.6 Add DVD

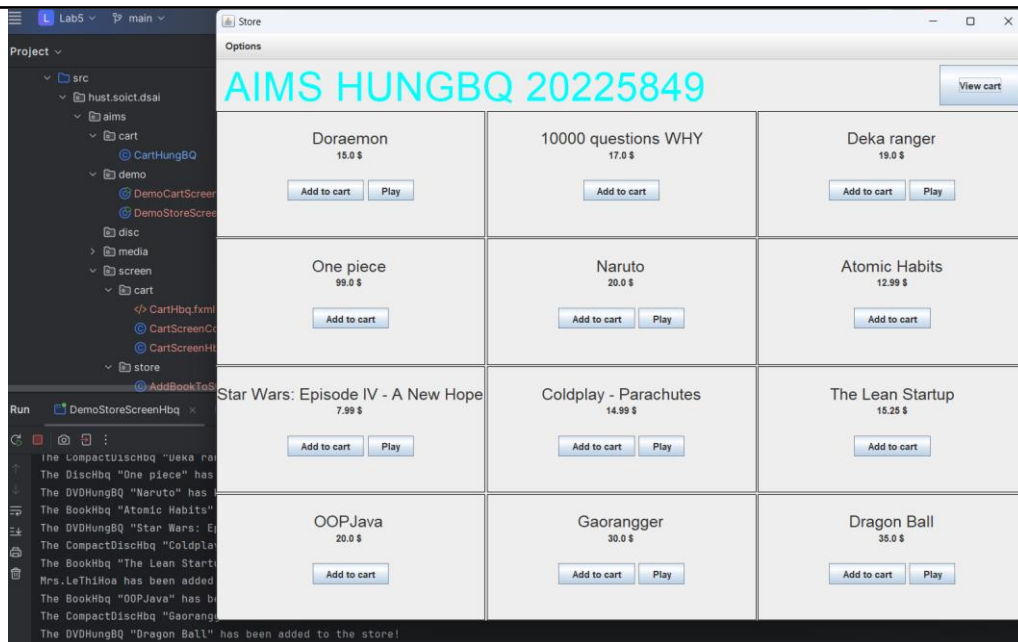


Figure 8.7 Store after add DVD

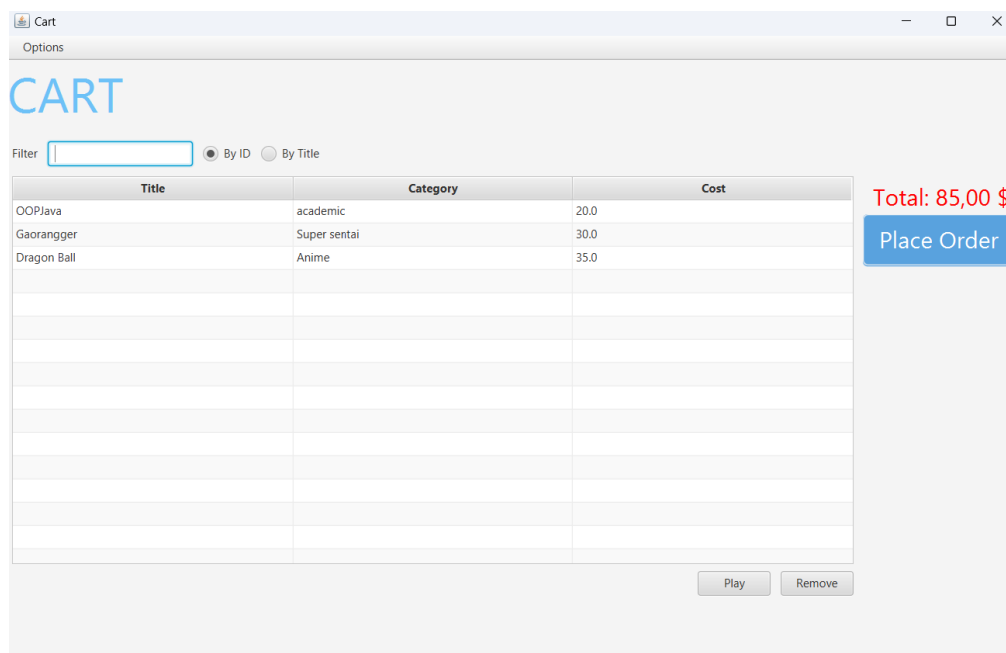


Figure 8.8 Cart

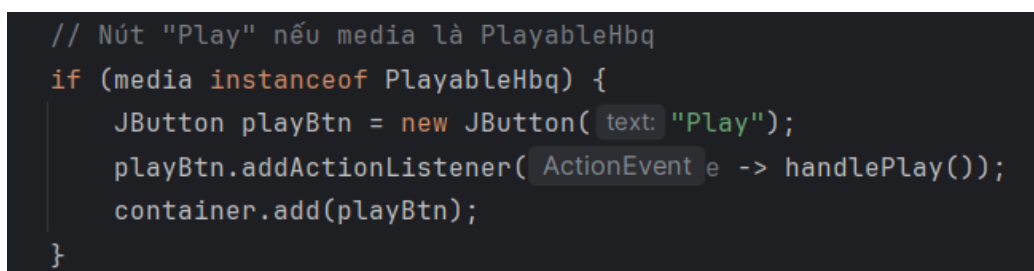


Figure 8.9 Exception

- Giải thích về trường hợp Exception:

- Đoạn code trên chỉ ra rằng khi đối tượng media là thể hiện của lớp Playable(hoặc bất kì lớp con nào của nó như DVD, CompactDisc hay Book), thì một nút "Play" sẽ được tạo và thêm vào giao diện.
- Có nghĩa là khi khởi tạo một đối tượng Book và thêm vào giỏ hàng thì giao diện của media đó sẽ không hiển thị nút “Play” còn đối tượng CompactDisc hay DVD thì ngược lại.

IX. UseCase Diagram

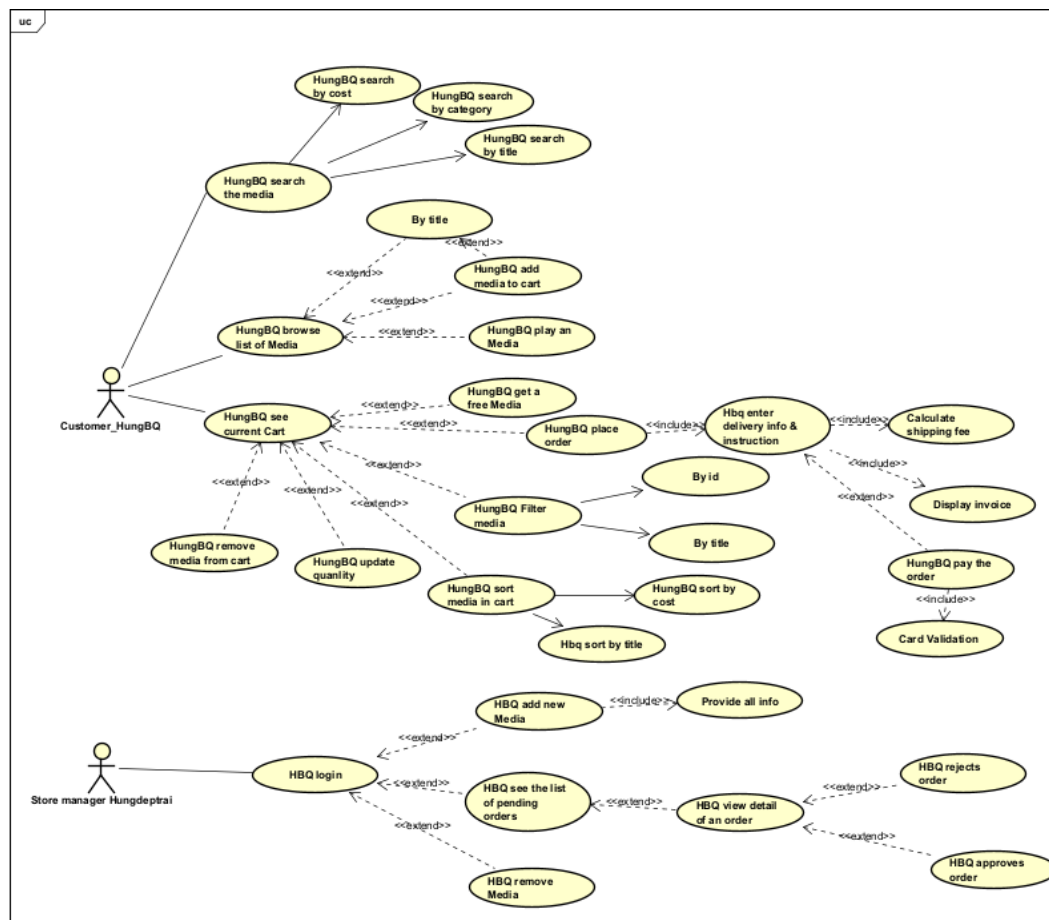


Figure 9.1 Usecase Diagram

```

classDiagram
    class Screen {
        +AddCDToStoreScreen
        +AddDVDToStoreScreen
        +AddBookToStoreScreen
        +AddBookToStoreScreen
        +StoreAddItemScreen
        +CartScreen
    }
    class Store {
        +MediaStore
        +StoreScreen
        +StoreHQ
    }
    class Media {
        +MediaHQ
        +BookHQ
        +DiscHQ
        +TrackHQ
        +CompactDiscHQ
        +DVDHQ
    }
    class Cart {
        +CartScreen
        +CartScreenController
    }
    class Exception {
        +Exception
        +PlayerException
        +AimHQ
    }

    Screen --> Store
    Screen --> Media
    Screen --> Cart
    Store --> Media
    Store --> Cart
    Media --> Cart
    Media --> Exception
    Cart --> Exception
    
```

The diagram illustrates the structure of a media management system. It features several main components: **Screen**, **Store**, **Media**, **Cart**, and **Exception**. The **Screen** component includes classes for adding items to the store, adding items to the store, adding items to the store, adding items to the store, adding items to the store, and a **CartScreen**. The **Store** component includes a **MediaStore**, a **StoreScreen**, and a **StoreHQ**. The **Media** component includes a **MediaHQ**, a **BookHQ**, a **DiscHQ**, a **TrackHQ**, a **CompactDiscHQ**, and a **DVDHQ**. The **Cart** component includes a **CartScreen** and a **CartScreenController**. The **Exception** component includes an **Exception**, a **PlayerException**, and an **AimHQ**. The diagram shows relationships between these classes, including inheritance and associations.

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